Lecture 2: Empirical Asset Pricing 1[1]

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[1] mainly based on Kelly et al. [2023], Gu et al. [2020]

Expected return

$$\mathrm{E}\left[R_{i,t+1} \mid \mathcal{I}_t\right] = \beta_{i,t}\lambda_t. \tag{1.2}$$

- $\mathbb{E}[R_{i,t+1} \mid \mathcal{I}_t]$ - Expected return of asset i at time $t+1$, given information $\mathcal{I}_t$.
- $\beta_{i,t}$ - Asset's exposure to systematic risk factors (time-varying risk loading).
- λ_t - Price of risk (compensation for bearing risk at time t).
- $\Rightarrow$ Expected return depends on risk exposure and risk premium.
- $\Rightarrow$ Generalizes CAPM/FF3 by allowing time-varying risk and return dynamics.

Capital Asset Pricing Model (CAPM)

- The CAPM formula relates the expected return of an asset to its risk relative to the market.
- The formula is:

$$E(R_i - R_f) = \beta_i (E(R_m) - R_f)$$

- Where:
  - $E(R_i)$ = Expected return of asset i
  - R_f = Risk-free rate
  - β_i = Beta of asset i, the sensitivity of the asset returns to market return
    - $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$
  - $E(R_m)$ = Expected return of the market

Criticisms of CAPM

- **1. Empirical Failures (The "Anomalies"):**
  - Low Explanatory Power: Low R^2 in cross-sectional regressions; many "alphas" remain unexplained.
  - Single Factor Insufficiency: Market beta alone cannot explain the cross-section of returns (e.g., Size, Value, Momentum effects).
- **2. Unrealistic Assumptions:**
  - Homogeneous Expectations: In reality, investors have heterogeneous beliefs and information sets ($\mathcal{I}_t^{Retail} \neq \mathcal{I}_t^{Inst}$).
  - Representative Agent: Everyone holds the market portfolio. In reality, portfolios are diverse.
- **3. Structural Limitation:**
  - Linearity Assumption: Classic CAPM assumes linear relation and static β_i.
- CAPM fails not because it is wrong, but because it is too restrictive.

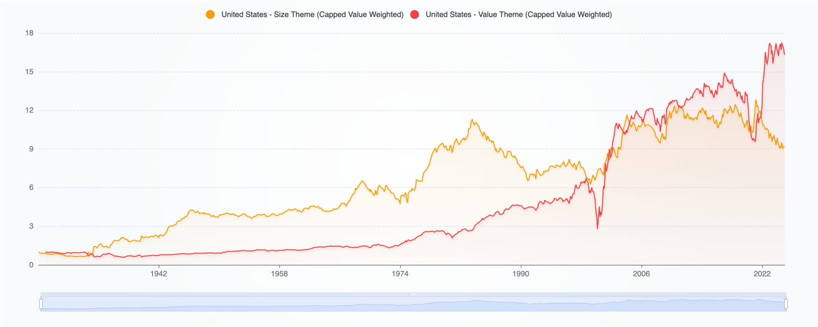
Fama–French Three-Factor Model

- The Fama–French model extends CAPM by allowing multiple sources of systematic risk.
- A reduced-form factor pricing model
- The expected return representation:

$$\mathbb{E}(R_i - R_f) = \beta_{i,M}\mathbb{E}(R_m - R_f) + \beta_{i,SMB}\mathbb{E}(\text{SMB}) + \beta_{i,HML}\mathbb{E}(\text{HML})$$

- **SMB (Small Minus Big):**
  - Factor return capturing size-related risk.
  - Small-cap stocks historically earn higher average returns than large-cap stocks.
- **HML (High Minus Low):**
  - Measures the return difference between value (high book-to-market) and growth (low book-to-market) stocks.
  - Value stocks historically earn higher average returns than growth stocks.

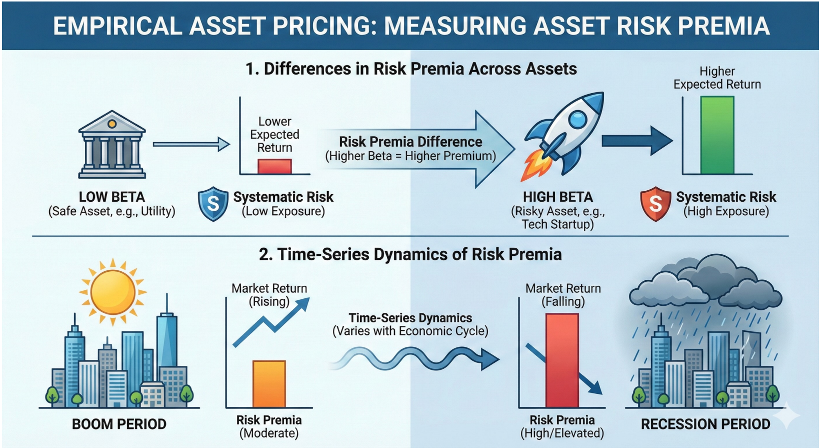
Visualization



- Positive long-run risk premia
- Factor performance is highly time-varying, with long periods of underperformance.

Introduction to Empirical Asset Pricing

- Empirical asset pricing is about measuring asset risk premia.
  1. Differences in risk premia across assets
     - Riskier assets (higher beta) have higher expected returns due to greater exposure to systematic risk.
  2. Time-series dynamics of risk premia
     - boom vs. recession periods

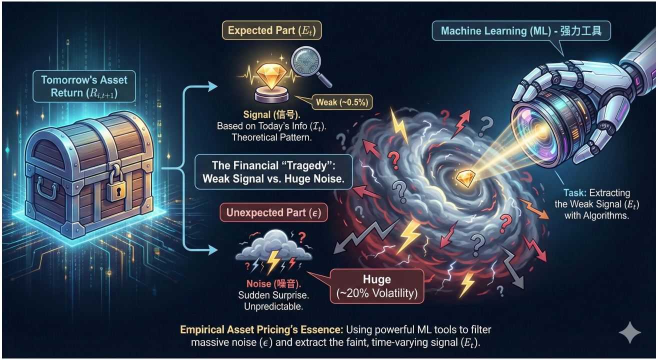


Introduction to Empirical Asset Pricing

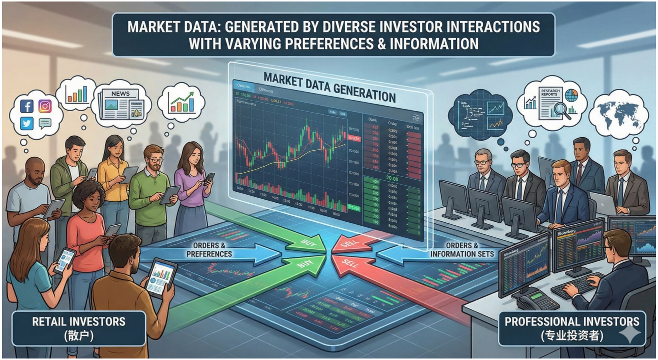
Definition: A return prediction measures an asset's conditional expected excess return:

$$R_{i,t+1} = E_t[R_{i,t+1}] + \epsilon_{i,t+1}$$

- Expectation component represents forecast conditional on information set $\mathcal{I}_t$
- $\epsilon_{i,t+1}$ captures unpredictable variation in returns



Market Complexity and Information Sets



- Prices are determined dynamically through trades, incorporating:
  - Public and private information
  - Obvious and subtle signals
  - Unambiguous and dubious inputs
- The conditional expectation of returns ($E_t[R_{i,t+1}]$) must capture these complexities.

Time Series vs. Cross Sectional Prediction

Two key strands in financial machine learning literature:

1. Time Series Models: Focus on aggregate asset returns like equity or bond index portfolios.
    - Developed earlier due to data availability.
    - Predictive regression techniques to capture time-varying discount rates.
    - Appear more in the ML community. Later this course.
2. **Cross Sectional Models:** Predict returns for many individual assets simultaneously.
    - Rapidly growing literature due to rich datasets. **Thousands of CS stocks**
    - Explains both time-variation and cross-sectional heterogeneity.
    - Enables empirical researchers to document new economic patterns.
    - More in the finance community.

Functional Representation of Expected Returns

Kelly et al. [2023]: Identify a function $g^*(z_{i,t})$ that fully explains expected returns across assets and time:

$$E_t[R_{i,t+1}] = g^*(z_{i,t}) \tag{1}$$

Improvements:

- P-dimensional predictor variables $z_{i,t}$
  - weak signal-to-noise ratio
- Estimation leverages panel data for stability
  - small data
- Contrast with standard asset pricing models which re-estimate cross-sectional models periodically
  - model update

Unsolved Problems / Challenges

1. **Heterogeneity Assumption:** Assumes a universal function $g^*(\cdot)$ governs all assets, potentially ignoring idiosyncratic structural differences.
2. **Information Asymmetry:** Researchers observe a subset $\mathcal{I}_{Model} \subset \mathcal{I}_{Market}$. Key pricing signals may be unobservable (omitted variable bias).
3. **Non-Stationarity (Distribution Shift):** Market dynamics constantly evolve due to regulation, technology, and arbitrage. History may not predict the future.

Data

- Many studies use CRSP-Compustat data for US stock returns and characteristics.
- Most economically important patterns in asset markets unfold over **monthly or lower** frequencies.
- Different researchers employ different signal sets:
  - Lewellen [2014]: 15 signals
  - Freyberger et al. [2020]: 36 signals
  - Gu et al. [2020]: 94 signals
- **Recent efforts have standardized these datasets**:
  - Jensen et al. [2023]: 153 stock signals available globally jkpfactors.com
  - Chen and Zimmermann [2021] provide US stock data openassetpricing.com
- *Aggregate* market return predictors from Welch and Goyal [2008], Rapach and Zhou [2022].

Data & Preprocessing 1

	permno	DATE	mvel1	RET	prc	SHROUT	beta	betasq	...	maxret	retvol	std_dolvol	std_turn	zerotrade	sic2	bm	bm_ia
0	10006	19260130	6.540000e+04	0.032732	110.250000	600	NaN	NaN	...	NaN	NaN	NaN	NaN	6.562500e-05	NaN	NaN	NaN
1	10022	19260130	1.120000e+04	0.017857	57.000000	200	NaN	NaN	...	NaN	NaN	NaN	NaN	2.573529e-06	NaN	NaN	NaN
2	10030	19260130	2.340000e+04	0.161667	174.250000	156	NaN	NaN	...	NaN	NaN	NaN	NaN	2.275000e-05	NaN	NaN	NaN
3	10049	19260130	1.850000e+04	0.141892	83.500000	250	NaN	NaN	...	NaN	NaN	NaN	NaN	2.187500e-05	NaN	NaN	NaN
4	10057	19260130	6.125000e+03	-0.035714	11.812500	500	NaN	NaN	...	NaN	NaN	NaN	NaN	9.943182e-06	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	..	...
4345503	93423	20201231	2.611343e+06	0.109665	34.099998	84977	1.921340	3.691548	...	0.176471	0.050187	0.519033	17.649093	3.803709e-09	79.0	NaN	NaN
4345504	93426	20201231	3.671460e+05	0.076239	31.480000	12552	1.301940	1.695047	...	0.159100	0.043004	0.473367	2.144264	3.236729e-08	36.0	NaN	NaN
4345505	93427	20201231	2.523235e+06	0.135851	77.589996	36938	0.847169	0.717695	...	0.048227	0.020206	0.294512	1.469369	2.227010e-08	36.0	NaN	NaN
4345506	93434	20201231	8.738802e+04	0.122605	2.930000	33482	0.389812	0.151954	...	0.116667	0.035277	0.935967	0.897075	1.037670e-07	1.0	NaN	NaN
4345507	93436	20201231	1.735365e+08	0.243252	705.669983	947901	1.349505	1.821163	...	0.101968	0.039271	0.531632	18.110463	2.654462e-09	37.0	NaN	NaN

[4345508 rows x 101 columns]

Figure: Data in Gu et al. [2020]

Preprocessing Avoid forward-look ahead bias!

Missing Data

Problem: Data gaps due to market entry (IPOs), exit (delisting/bankruptcy), or reporting delays.

- **Cross-sectional Imputation:** Replace missing values with the **median** at time t. basicly we use Cross-sectional Imputation
- **Time-series Imputation:** Forward fill using the most recent valid observation (Carried Forward).

Data & Preprocessing 2

Normalization and Outliers

Problem: Extreme values distort model training and optimization.

- **Rank-Normalization (Recommended):** Map raw values to rankings in [-1, 1].

$$x_{i,t}^{rank} = \frac{2 \cdot \text{Rank}(x_{i,t})}{N_t + 1} - 1$$

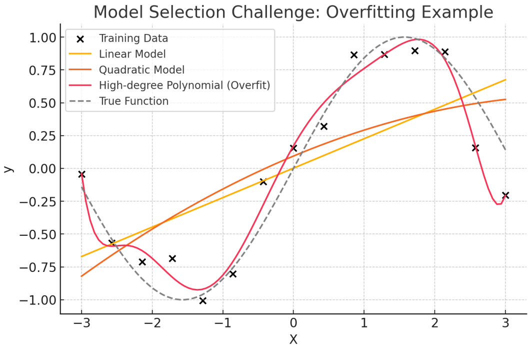
- **Standardization (Z-Score):** $z_{i,t} = (x_{i,t} - \mu_t)/\sigma_t$. (Less robust to outliers).

- Preprocessing is not a purely technical choice.
- Imply how investors process information.
  - Rank-normalization $\rightarrow$ investors care about relative rankings.
  - Standardization $\rightarrow$ investors respond to standardized signals.
  - Carry-forward missing values $\rightarrow$ investors use last observed information.

Model Selection Challenges

- Estimating and selecting models is central to machine learning.
- In-sample performance exaggerates predictive power due to overfitting.
- Large models fit training data perfectly but may not generalize well.
- Small models or Large models+Regularization?

Model Selection Challenges



- Linear regression (simple, interpretable, but may underfit).
- High-degree polynomial regression (fits training data perfectly but risks overfitting).
- bias-variance tradeoff: complex models can memorize noise rather than learning patterns.
- In finance, overfitting is dangerous precisely because the true function is unobserved and signals are weak.

Model Selection Methods

- **Information Criteria:** Penalizes model complexity

Criterion	Penalty Term	Model Complexity	Assumption on True Model
AIC	$2P$	More complex models	No assumption
BIC	$P\log(N)$	Simpler models	true model is among candidates

Table: Key Differences Between AIC and BIC

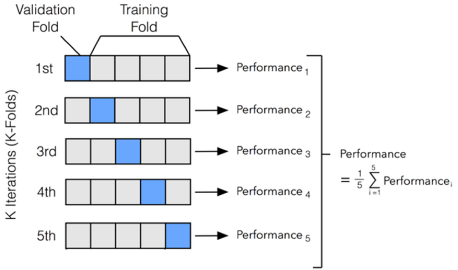
AIC and BIC in Financial ML

- P denotes the number of free parameters; N is the sample size.
- AIC/BIC are designed for low-dimensional parametric models.
- In financial ML, model complexity (including non-linearity, interactions, regularization) is not well captured by parameter counts.
- Asset pricing models are inherently misspecified, violating key assumptions of BIC.
- As a result, AIC and BIC are rarely used for model selection in modern financial ML.

Cross-Validation

Data-driven approach evaluating "pseudo" out-of-sample performance

- Information criteria rely on theoretical assumptions
- Cross-validation is flexible but computationally expensive



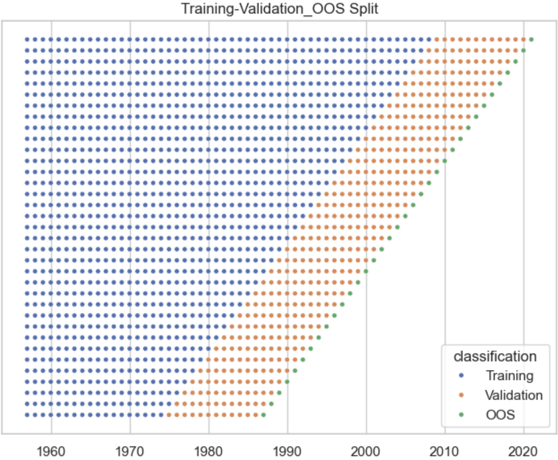
Sample Splitting Design

- Dataset split into three disjoint subsamples:
    1. **Training sample:** Used to estimate candidate models.
    2. **Validation sample:** Used for hyperparameter tuning.
    3. **Testing sample:** Final evaluation on unseen data.
- Ensures model tuning does not influence test performance.
- **K-fold cross-validation** for uncorrelated data.
    - cross-sectional stock returns at a given time

Time Series Considerations

- Temporal ordering must be preserved to prevent data leakage.

Recursive Time-Ordered Cross-Validation



- Uses fixed/expanding training windows for sequential learning.
- Adaptively updates models at each time step.
- Allows model specification to evolve as new data becomes available.

Simple Linear Model for Stock Returns

The predictive benchmark for stock returns:

$$R_{i,t+1} = \beta' z_{i,t} + \epsilon_{i,t+1}.$$

Predictive function:

$$g(z_{i,t}) = \beta' z_{i,t}.$$

泰勒展开

- $z_{i,t}$ represents stock-level predictive features of dimension P.
- Linear model as a reasonable benchmark for comparison against machine learning methods.

Solution of OLS? $Y = X\beta + \epsilon$

Gauss-Markov Thm

Assumptions:

- Linearity
- Independence (No serial correlation)
- Homoscedasticity (Constant variance)

Under some conditions ($E[\epsilon_{t+1} \mid z_t] = 0, \ldots$)

- - OLS is the Best Linear Unbiased Estimator (BLUE) within the class of linear unbiased estimators.
- - There exist biased estimators with lower variance (e.g., James-Stein, Ridge)
- **But in ML, we care about MSE, not Unbiasedness.**

$$\text{MSE} = \text{Variance} + \text{Bias}^2$$

The Curse of Dimensionality (Large P):

$$\text{Variance of Prediction} \propto \sigma^2 \frac{P}{NT}$$

- When P is large, OLS variance explodes.
- **Solution:** Use Biased estimators (Ridge/Lasso) to reduce

Derivation

Step 1: Linear model and OLS estimator

$$Y = X\beta + \varepsilon, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top Y = \beta + (X^\top X)^{-1} X^\top \varepsilon.$$

Assume

$$\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I).$$

Step 2: Distribution of OLS prediction

For a new observation with features $z_{i,t}$,

$$\hat{R}_{i,t+1} = z_{i,t}^\top \hat{\beta}.$$

Conditioning on X,

$$\hat{R}_{i,t+1} \mid X \sim \mathcal{N}\left(z_{i,t}^\top \beta,\ \sigma^2\, z_{i,t}^\top (X^\top X)^{-1} z_{i,t}\right).$$

Step 3: Large-sample approximation

If $\frac{1}{NT} X^\top X \approx \Sigma_z$, then

$$\mathrm{Var}(\hat{R}_{i,t+1}) \approx \frac{\sigma^2}{NT}\, z_{i,t}^\top \Sigma_z^{-1} z_{i,t}.$$

Step 4: Order-of-magnitude intuition

If features are standardized so that $\Sigma_z \approx I$ and $\|z_{i,t}\|^2 \approx cP$, then

$$\mathrm{Var}(\hat{R}_{i,t+1}) \approx c\sigma^2 \frac{P}{NT}.$$

Related Literature

- Haugen and Baker [1996], Lewellen [2014] and Gu et al. [2020] employed large sets of predictive signals.
- They recursively trained linear panel models and assessed out-of-sample performance.
- Evaluations were conducted in economic terms via trading strategy performance.
- This contrasts with traditional portfolio sorting, which assumes fixed return relationships.
  - sorts stocks into bins based on one or two characteristics
  - essentially a “zero-parameter” return prediction model that asserts a functional form for g without performing estimation

		Factor B [Fb]		
		Low	Mid	High
Factor A [Fa]	Low	P1	P2	P3
	Mid	P4	P5	P6
	High	P7	P8	P9

Empirical Findings of Linear Models

Haugen and Baker [1996]

- Large number of signals (40 continuous variables, sector dummies)
- International data analysis
- Showed return forecasts from linear models outperform the aggregate market.

Lewellen [2014]

- 15 continuous variables
- Out-of-sample $R^2 \approx 1\%$ per month
- Demonstrated strong trading strategy performance with a Sharpe ratio of 1.72.

Gu et al. [2020]

- 94 characteristics for each stock, interactions of each characteristic with 8 macro indicators, and 74 industry sector dummy variables
- totaling more than 900 baseline signals.
- OLS using all predictors produces an OOS R^2 of -3.46%.

Portfolio Performance

	Equal-weighted					Value-weighted				
	Pred	Avg	Std	t-stat	SR	Pred	Avg	Std	t-stat	SR
Low (L)	-0.90	-0.32	7.19	-0.84	-0.15	-0.76	0.11	6.01	0.37	0.06
2	-0.11	0.40	5.84	1.30	0.24	-0.10	0.45	4.77	1.89	0.32
3	0.21	0.60	5.46	2.06	0.38	0.21	0.65	4.65	2.84	0.49
4	0.44	0.78	5.28	2.74	0.51	0.44	0.69	4.67	2.97	0.51
5	0.64	0.81	5.36	2.82	0.52	0.63	0.81	5.01	3.34	0.56
6	0.83	1.04	5.36	3.62	0.67	0.82	0.88	5.22	3.28	0.58
7	1.02	1.12	5.55	3.68	0.70	1.01	1.04	5.67	3.46	0.64
8	1.25	1.31	5.97	4.04	0.76	1.24	1.15	6.03	3.62	0.66
9	1.55	1.66	6.76	4.38	0.85	1.54	1.34	6.68	3.80	0.69
High (H)	2.29	2.17	7.97	4.82	0.94	2.19	1.66	8.28	3.73	0.70
H - L	3.20	2.49	5.02	10.00	1.72	2.94	1.55	6.56	4.51	0.82

Table: from Lewellen [2014]

The Problem with Traditional OLS Estimation

- When the number of predictors P is very large, OLS becomes statistically unstable and overfits noise.
  - Gu et al. [2020] show that traditional OLS estimation fails with too many predictors.
  - Estimates tend to have low in-sample bias but large variance
  - In return prediction, the signal-to-noise ratio is low, making OLS prone to errors.
- Solutions?

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- Solutions?
  - **1. Subset Selection:** Assumes only *a small subset with strong effects* matters (Sparsity). Hard drops the rest.
  - **2. Regularization:** Keeps all variables but shrinks coefficients to control complexity (Bias-Variance Tradeoff).

Subset Selection

General Idea:

- Search across all permutations of models
- Choose the best model according to some criterion

Challenges:

- Set of models may be large
- Even restricting to models of size k requires searching over $\binom{P}{k}$ candidates.
- Combinations are:

$$\sum_{k \leq P} \binom{P}{k} = 2^P$$

Need Algorithms!

Forward/Backward Stepwise

Forward Stepwise:

- Starts with the intercept
- Sequentially adds the predictor that most improve the fit
- Delivers a sequence of selected models for each subset size k

Backward Stepwise:

- Starts with all predictors
- Sequentially drops the predictor with the least impact on fit
- $\times$ for high-dimensional finance data

Greedy, makes the locally optimal choice at each step with the hope that these local solutions will lead to a globally optimal solution

Forward/Backward Stepwise

- Can use both, and choose best of the two
- within ACCEPTABLE computing time
- Cannot use standard test statistics for selected model, must account for the selection process/multiple tests
  - Run 100 separate regressions, each using one predictor. Select the variable with the largest t-statistic / lowest p-value.
  - Even if all predictors are pure noise: With 100 tests, it is very likely that at least one looks statistically significant at 5%.
  - This is not evidence of predictability — it is selection by chance.
  - Solutions?

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  - Solutions?
  - **Cross-Validation**
  - Regularization, Statistical Adjustment (for reference)

Regularization

Optimization: $\min_\beta \underbrace{\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(R_{i,t+1}-\beta' z_{i,t}\right)^2}_{RSS} + Penalty$

Lasso (L_1 penalty): $\lambda \sum_{i=1}^{P} |\beta_i|$

Ridge (L_2 penalty): $\lambda \sum_{i=1}^{P} \beta_i^2$

Pros and cons, why, their applications in Finance?

- high-dim, noise, market dynamics, efficiency?

LASSO

- \+ Forces some coefficients to be exactly zero.
- \+ Automatic feature selection, encouraging sparsity.
- \+ Useful when you expect a subset of predictors to be important.
  - consistent (in terms of selecting the correct set of predictors) under certain conditions

- \- Unstable, for highly correlated predictors.
- \- Might discard variables that are important but weakly correlated with the response.
- \- Numerical methods.
  - Non-differentiable L_1 regularization term
  - Coordinate Descent: optimize each coefficient one at a time, holding the others fixed.
  - Least Angle Regression (LARS)

Ridge

- $+$ Shrinks coefficients towards zero.
- $+$ Suitable for situations where most predictors contribute to the outcome.
- $+$ Closed-form solution.
- $+$ Stable in the presence of multicollinearity.

Ridge

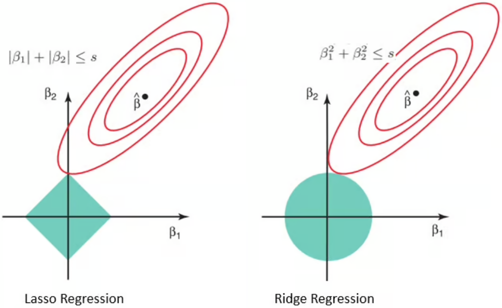
- + Shrinks coefficients towards zero.
- + Suitable for situations where most predictors contribute to the outcome.
- + Closed-form solution.
- + Stable in the presence of multicollinearity.
    - Small changes in the data can lead to large variations in the estimated coefficients of OLS.
    - Design matrix $X^T X$ becomes ill-conditioned.
    - Ridge adds λI to the diagonal, improving the conditioning of the matrix and making inversion more stable.
    - By shrinking the coefficients, Ridge reduces the sensitivity of the model to small fluctuations
- - Does not perform feature selection.
- - Can be harder to interpret.
- - Ridge not invariant to scale, so normally standardize z first

Visualization

The constrained form of Lasso is:

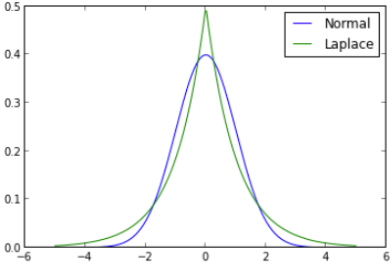
$$\min_{\beta} \left(\frac{1}{2n} \|\mathbf{y} - X\beta\|_2^2 \right), s.t. \quad \|\beta\|_1 \le t$$

Here, t is a threshold related to λ.



- RSS has elliptical contours centered at OLS
- L_1 constraint has "sharp edges" giving high probability of tangency at the corner of a hypercube

Bayesian view



Bayesian Interpretation:

- Ridge $\leftrightarrow$ normal prior
- Lasso corresponds to Laplace prior

Lasso Interpretation:

- Laplace prior sharply peaked at zero, concentrating probability mass around zero
- The distribution decays exponentially as β moves away from zero.

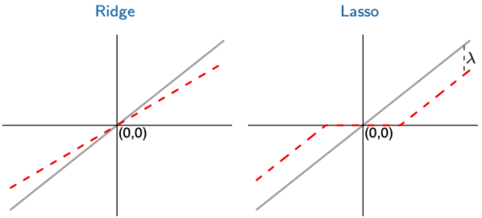
Ridge vs Lasso

Estimator	Formula
Ridge	$\hat{\beta}_j^{\mathrm{OLS}}/(1+\lambda)$
Lasso	$\mathrm{sign}\left(\hat{\beta}_j^{\mathrm{OLS}}\right)\left(\left\|\hat{\beta}_j^{\mathrm{OLS}}\right\|-\lambda\right)^+$

Table: For $X'X = \mathbf{I}$.

Think about Why?

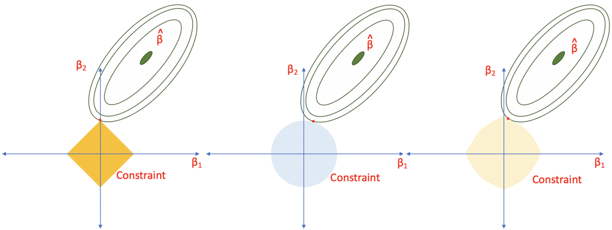
- **Ridge:** Performs proportional shrinkage on the coefficients
- **LASSO:** translates each coefficient by constant factor λ, truncating at zero (Soft thresholding)



The Elastic Net Loss Function

$$\mathcal{L}(\beta;\rho,\lambda) = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(R_{i,t+1} - \beta' z_{i,t}\right)^2 + \lambda(1-\rho)\sum_{j=1}^{P}|\beta_j| + \frac{1}{2}\lambda\rho\sum_{j=1}^{P}\beta_j^2$$

- Key hyperparameters: λ and ρ.
- Elastic net allows for a balance between Ridge and LASSO regularization.



Performance of ENet Portfolios

	Equal-weighted				Value-weighted			
	Pred	Avg	Std	SR	Pred	Avg	Std	SR
Low(L)	-0.04	-0.24	6.43	-0.13	-0.04	0.24	5.44	0.15
2	0.27	0.44	5.90	0.26	0.27	0.56	4.84	0.40
3	0.44	0.52	5.27	0.34	0.44	0.53	4.50	0.40
4	0.59	0.70	4.73	0.51	0.59	0.72	4.11	0.61
5	0.73	0.71	4.94	0.49	0.73	0.72	4.42	0.57
6	0.87	0.79	5.00	0.55	0.87	0.85	4.60	0.64
7	1.01	0.85	5.21	0.56	1.01	0.87	4.75	0.64
8	1.17	0.88	5.47	0.56	1.16	0.88	5.20	0.59
9	1.36	0.85	5.90	0.50	1.36	0.80	5.61	0.50
High(H)	1.72	1.86	7.27	0.89	1.66	0.84	6.76	0.43
H-L	1.76	2.11	5.50	1.33	1.70	0.60	5.37	0.39

Table: from Gu et al. [2020]

Generalized Additive Models (GAMs)

The general form of a GAM is:

$$g(\mathbb{E}[Y|\mathbf{X}]) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p)$$

Where:

- Y: Response variable.
- $\mathbf{X} = (X_1, X_2, \ldots, X_Q)$: Predictor variables.
- $g(\cdot)$: Link function.
- $f_1, f_2, \ldots, f_Q$: Smooth functions of the predictors.

Pros and Cons

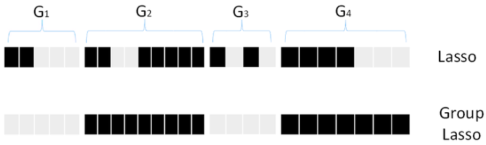
- $+$ Non-linear relationships
- $+$ Interpretability
- \- Overfitting
- \- Interactions

Group Lasso Penalty in GAM

Freyberger et al. [2020] combine Group Lasso with GAM to predict stock-month returns.

$$\lambda \sum_{j=1}^{P} \left(\sum_{k=1}^{K} \tilde{\beta}_{k,j}^2 \right)^{1/2},$$

where $\tilde{\beta}_{k,j}$ is the coefficient on the k^{th} basis function applied to stock signal j. Group lasso selects either all K transformations associated with a given characteristic j, or none of them.

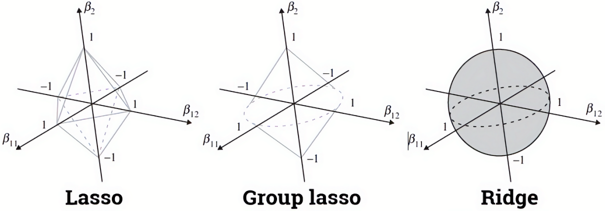


Group Lasso Explanation

- when $K = 1$ (each predictor is in its own group), ?
- when $P = 1$ (only one single group), ?
- Think about what Group LASSO looks like and why?

Group Lasso Explanation

- when $K = 1$ (each predictor is in its own group), ?
- when $P = 1$ (only one single group), ?
- Think about what Group LASSO looks like and why?
- Group sparsity
- Choice of groupings



Performance of Group LASSO

Freyberger et al. [2020] show that

- Less than half of the commonly studied stock signals in the literature have independent predictive power for returns
- The full nonlinear specification dominates the nested linear specification in out-of-sample performance.

Table A.1: **Selected Characteristics in Nonparametric Model**

This table reports the selected characteristics from the universe of thirty-six firm characteristics we discuss in Section IV for different numbers of knots, the sample size, and in-sample Sharpe ratios of an equally-weighted hedge portfolio going long stocks with the highest predicted returns and shorting stocks with lowest predicted returns. q indicates the size percentile of NYSE firms. The sample period is July 1963 to June 2014 unless otherwise specified.

Firms		*Size* $> q_{10}$	*Size* $> q_{10}$	*Size* $> q_{20}$	*Size* $> q_{50}$	All	All	*Size* $> q_{10}$
Sample		Full	Full	Full	Full	1963-1990	1991-2014	1991-2014
Knots		14	4	4	9	4	4	9
Sample Size		764,475	764,475	607,416	318,674	416,980	656,080	462,321
# Selected		9	15	13	4	14	23	8
Sharpe Ratio		2.04	2.21	2.08	1.45	2.89	3.01	1.89
Characteristic	# Selected	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A2ME	2					A2ME	A2ME	
AT	1						AT	
Beta	2					Beta	Beta	
C	1						C	
CTO	1						CTO	
D2A	2		D2A				D2A	
E2P	5	E2P	E2P	E2P			E2P	E2P
FC2Y	2		FC2Y				FC2Y	
Free CF	1						Free CF	
Idio vol	2					Idio vol	Idio vol	
Investment	2			Investment		Investment		
LME	6	LME	LME	LME		LME	LME	LME
Lturnover	6	Lturnover	Lturnover	Lturnover		Lturnover	Lturnover	Lturnover
NOA	1		NOA					
OA	3		OA	OA		OA		
PCM	2		PCM	PCM				
PM	1						PM	
Profitability	1						Profitability	
Rel to High	6	Rel to High	Rel to High	Rel to High		Rel to High	Rel to High	Rel to High
RNA	2		RNA				RNA	
r_{12-2}	6	r_{12-2}	r_{12-2}	r_{12-2}	r_{12-2}	r_{12-2}	r_{12-2}	
r_{12-7}	3		r_{12-7}	r_{12-7}			r_{12-7}	
r_{2-1}	7	r_{2-1}	r_{2-1}	r_{2-1}	r_{2-1}	r_{2-1}	r_{2-1}	r_{2-1}
r_{36-13}	6	r_{36-13}	r_{36-13}	r_{36-13}		r_{36-13}	r_{36-13}	r_{36-13}
SGA2M	7	SGA2M	SGA2M	SGA2M	SGA2M	SGA2M	SGA2M	SGA2M
Spread	2					Spread	Spread	
SUV	7	SUV	SUV	SUV	SUV	SUV	SUV	SUV

Never selected: ATO, BEME, DPI2A, Lev, OL, Q, ROA, ROE, S2P

Motivation

- Regularization is beneficial for high-dimensional problems as it reduces degrees of freedom.
- But may produce suboptimal forecasts when predictors are highly correlated.
- Example:
  - Each predictor is equal to the forecast target plus some i.i.d. noise term.

Motivation

- Regularization is beneficial for high-dimensional problems as it reduces degrees of freedom.
- But may produce suboptimal forecasts when predictors are highly correlated.
- Example:
  - Each predictor is equal to the forecast target plus some i.i.d. noise term.
  - The sensible forecasting solution is to simply use the average of predictors in a univariate predictive regression.
- **Dimension reduction**: Create linear combinations of predictors to REDUCE NOISE AND ISOLATE SIGNAL.
- Common techniques: Principal Components Regression (PCR), Partial Least Squares (PLS), and extensions.

Common Idea of Dimension Reduction

Consider the predictive regression model:

$$y_{t+1} = x_t' \theta + \epsilon_{t+1}$$

- y_{t+1} refers to future variables (e.g., market returns).
- x_t is a $P \times 1$ vector of predictors.
- Dimension reduction replaces high-dimensional predictors x_t with low-dimensional hidden "factors" f_t, which summarize the useful information in x_t.
- The factor model relationship:

$$x_t = \underbrace{\beta}_{P \times K} \underbrace{f_t}_{K \times 1} + u_t$$

- Rewriting the predictive regression using factors:

$$y_{t+1} = f_t' \underbrace{\gamma}_{K \times 1} + \tilde{\epsilon}_{t+1}$$

Principal Components Regression (PCR)

Principal Component Analysis (PCA) + linear regression

- **Step 1**: Combine predictors into a few linear combinations $\widehat{f}_t = \Omega_K x_t$.
- Find the combination weights Ω_K recursively:

$$w_j = \arg\max_{w} \widehat{\text{Var}}\left(x_t' w\right)$$

$$\text{s.t.} \quad w' w = 1, \quad \widehat{\text{Cov}}\left(x_t' w, x_t' w_l\right) = 0, \quad l = 1, 2, \ldots, j-1 \tag{3.12}$$

- Goal: The choice of components aims to best preserve the covariance structure among predictors.
- **Step 2**: Use the estimated components $\widehat{f}_t$ in a standard linear predictive regression.

What is the relation between PCA and Eigen-decomposition?

PCR analysis

Pros:

- Handles Multicollinearity: PCR uses orthogonal components
- Reduces Overfitting: selecting only a few principal components
- Interpretation: low-dimensional structures

Cons:

- Not Optimized for Prediction

PCA in Financial Data

- In asset pricing, first PC of covariance/correlation matrix looks like?

PCA in Financial Data

- In asset pricing, first PC of covariance/correlation matrix looks like?
- resemble a market factor, see Avellaneda and Lee [2010], Lettau and Pelger [2020], Avellaneda et al. [2022].

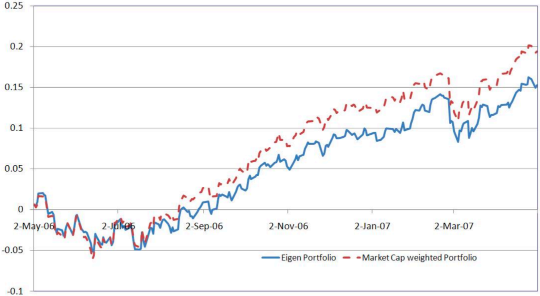


Figure 3: Comparative evolution of the principal eigenportfolio and the capitalization-weighted portfolio from May 2006 to April 2007. Both portfolios exhibit similar behavior.

Partial Least Squares (PLS)

- **PLS**: Focuses on reducing dimensionality based on correlation with forecast target.
- **Optimization**: Maximize covariance between predictors and target.
- PLS finds latent factors (also called components) that are:
  - linear combinations of predictors.
  - highly correlated with the target variable.

PLS Procedure

different from PCA, this method max the cov

- **Step 1: Find the First Latent Variable**:
  - Maximize the covariance between the first latent variable t_1 and the response y:

$$t_1 = Xw_1 \quad \text{where} \quad w_1 = \text{argmax}\left(\text{cov}(t_1, y)\right)$$

- **Step 2: Deflate the Predictor Matrix**:
  - Remove the contribution of t_1 from X:

$$X' = X - t_1 w_1^T$$

- **Step 3: Repeat for Subsequent Latent Variables**:
  - Find additional latent variables $t_2, t_3, \ldots$, each orthogonal to the previous ones.
- **Step 4: Regress the Response y on the Latent Variables**:
  - Regress y on the latent variables: $y = t_1c_1 + t_2c_2 + \cdots + t_pc_p + \epsilon$.

PCR vs PLS

- PCR: Reduces dimensions based on covariance among predictors.
- PLS: Reduces dimensions based on correlation with target (better for prediction).
- PLS avoids irrelevant components and adapts more closely to forecasting goals.

Value-weighted Portfolio Performance of PLS and PCR

	PLS				PCR			
	Pred	Avg	Std	SR	Pred	Avg	Std	SR
Low(L)	-0.83	0.29	5.31	0.19	-0.68	0.03	5.98	0.02
2	-0.21	0.55	4.96	0.38	-0.11	0.42	5.25	0.28
3	0.12	0.64	4.63	0.48	0.19	0.53	4.94	0.37
4	0.38	0.78	4.30	0.63	0.42	0.68	4.64	0.51
5	0.61	0.77	4.53	0.59	0.62	0.81	4.66	0.60
6	0.84	0.88	4.78	0.64	0.81	0.81	4.58	0.61
7	1.06	0.92	4.89	0.65	1.01	0.87	4.72	0.64
8	1.32	0.92	5.14	0.62	1.23	1.01	4.77	0.73
9	1.66	1.15	5.24	0.76	1.52	1.20	4.88	0.86
High(H)	2.25	1.30	5.85	0.77	2.02	1.25	5.60	0.77
H-L	3.09	1.02	4.88	0.72	2.70	1.22	4.82	0.88

Table: from Gu et al. [2020]

Single vs. Combined Forecasts

- Multiple forecasts are often available to decision makers.
- Should a single dominant forecast be used? Or a combination?

Forecast Combinations:

- Offer **diversification gains** by combining different forecasts.
- heterogeneous beliefs where investors are represented as different machine learning model specifications, (compare to analyst forecast disagreement) see Bali et al. [2023]
- However, combining only makes sense if:
  - One cannot identify ex ante a particular model that generates **smaller forecast errors** than competitors.
  - The forecast errors of the best model can be **hedged by other models' forecast errors**.

Any thoughts about combination ways?

Forecast Combination

Granger and Ramanathan [1984]

- Forecasts from M different people/models: $\hat{y}_{i,t+1}, i = 1, \ldots, M$

How to combine these forecasts into a "super" predictor?

- The combined forecast: $\hat{y}^C_{t+1}$
- Minimize Mean Squared Error (MSE): Project y_{t+1} onto all $\{\hat{y}_{i,t+1}\}_{i=1}^M$

Combination formula:

$$\hat{y}^C_{t+1} = \omega_0 + \sum_i \omega_i \hat{y}_{i,t+1}$$

If each $\hat{y}_{i,t+1}$ is unbiased, one might impose:

$$\omega_0 = 0 \quad \text{and} \quad \sum_i \omega_i = 1$$

1. Simple Combinations

- Uses central tendency of forecasts, e.g., $\omega_i = 1/N$
- For less outlier sensitivity:
  - Median: $\hat{y}^C_{t+1} = \text{median}\left(\hat{y}_{1,t+1}, \ldots, \hat{y}_{N,t+1}\right)$
  - Trimmed mean: Drop some from each tail, average remaining

2. Relative Performance Weights

- OLS combo weights suffer from imprecision in estimates of forecast error covariance matrix (Σ_ϵ)
- One simple answer: Ignore forecast error correlations
- Weight based on performance of each individual model:

$$\hat{y}^C_{t+1} = \sum_i \omega_i \hat{y}_{i,t+1}, \quad \omega_i = \frac{1/\mathrm{MSE}_i}{\sum_j 1/\mathrm{MSE}_j}$$

- Or weights based on model rank, less sensitive to outliers (Aiolfi and Timmermann 2006):

$$\omega_i = \frac{1/\mathcal{R}_i}{\sum_j 1/\mathcal{R}_j}, \quad \mathcal{R}_i \text{ is the MSE rank of } i$$

3. Shrinkage Combinations

- Diebold and Pauly [1990] derive this from model:

$$Y = \omega' \hat{Y} + \epsilon, \quad \epsilon \sim N(0, \sigma^2 \mathbf{I})$$

- Conjugate priors:

$$\omega \sim N\left(1/N, \frac{1-\lambda}{\lambda} \mathrm{Cov}[\hat{Y}]\right), \quad \sigma^2 \sim \mathrm{IG}(\bar{\sigma}^2, v)$$

- Can select λ via cross-validation, etc
- Shrink weights toward mean forecast

$$\omega_i = \lambda \hat{\omega}_i + (1-\lambda)(1/N)$$

4. Time-varying Combinations

- Individual models can be differently affected by structural breaks:
  - Some adapt quickly, others slowly
  - Difficult to detect instability in real-time
- Forecast combos – averaging models with different degrees of adaptability – effective for managing DGP instability

4. Time-varying Combinations

- Individual models can be differently affected by structural breaks:
  - Some adapt quickly, others slowly
  - Difficult to detect instability in real-time
- Forecast combos – averaging models with different degrees of adaptability – effective for managing DGP instability

Some approaches:

- Bates and Granger [1969]: Rolling MSE weights

$$\omega_{k,t} = \frac{1/E_{k,t}}{\sum_i 1/E_{i,t}},$$

  where $E_k = \sum_{\tau=t-\text{win}+1}^{t} e^2_{k,\tau}$
- Diebold and Pauly [1987]: Exponentially smoothed MSE weights

Empirical Results

Table 1: Individual and Combination Forecast Results

The table reports monthly out-of-sample results for 14 univariate predictive regression and two combination forecasts of the US market excess return. The out-of-sample period is 1957:01–2020:12. Each univariate predictive regression forecast is based on the predictor variable in the first column. C-Mean is a combination forecast based on the arithmetic mean of the individual univariate predictive regression forecasts. DMSFE is a combination forecast that attaches more weight to individual univariate predictive regression forecasts with lower MSFE over a holdout out-of-sample period. R^2_{OS} is the Campbell and Thompson (2008) out-of-sample R^2 statistic. It measures the proportional reduction in MSFE for the univariate predictive regression or combination forecast via-à-vis the prevailing mean benchmark. For the positive R^2_{OS} statistics, *, **, and *** indicate that the reduction in MSFE is significant at the 10%, 5%, or 1% level, respectively, according to the Clark and West (2007) test. Gain is the annualized increase in CER for a mean-variance investor with a relative risk aversion coefficient of five who uses the univariate predictive regression forecast or combination forecast instead of the prevailing mean benchmark to allocate between stocks and risk-free Treasury bills.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Overall		Expansion		Recession	
Predictor	R^2_{OS} (%)	Gain (%)	R^2_{OS} (%)	Gain (%)	R^2_{OS} (%)	Gain (%)
log(DP)	−0.36	0.32	−1.53	−1.84	2.29***	12.51
log(DY)	−0.75	0.46	−2.55	−2.42	3.34***	16.85
log(EP)	−1.92	0.24	−2.29	−0.76	−1.05	5.69
log(DE)	−1.75	−0.41	−1.02	0.00	−3.41	−2.66
SVAR	−0.44	−0.19	0.03	−0.29	−1.50	0.45
BM	−1.93	−1.17	−2.79	−2.48	0.02	6.03
NTIS	−0.60	−0.05	0.71	0.88	−3.56	−5.49
TBL	0.21*	1.47	−0.41	0.49	1.62*	6.94
LTY	−0.83	1.15	−1.72	0.04	1.20	7.36
LTR	−0.08	0.49	−0.70	−0.14	1.33*	3.81
TMS	0.02	1.07	−0.42	−0.01	0.99*	7.13
DFY	−0.03	0.23	−0.06	−0.10	0.02	1.89
DFR	−0.07	0.74	0.35*	0.53	−1.03*	1.99
INFL	−0.03	0.36	0.15	0.20	−0.42	1.34
C-Mean	0.33**	1.04	0.11	0.31	0.84**	5.13
DMSFE	0.39**	1.27	0.08	0.30	1.10**	6.74

Figure: from Rapach and Zhou [2022]

Empirical Summary of Forecast Combination

1. Combining methods typically outperform individual forecasts in the panel, often by a wide margin.
2. Simple combining methods—e.g., mean, trimmed mean, median—often perform as well as or better than more sophisticated methods. – "forecast combining puzzle"

Homework 1

This assignment aims to assess your understanding of linear regression, including its strengths and limitations, while applying it to predict stock returns using U.S. stock market data.

- **Basic Requirements:**
  - Run linear models (e.g., OLS, LASSO, Ridge) using all firm characteristics to predict returns on U.S. stock market data.
    - GKX data Please don't distribute without my authorization.
    - cols_del = ['SHROUT','mve0','prc','permno','DATE','sic2']
  - Recursive Time-Ordered Cross-Validation
  - Evaluate the statistical performance (MSE, R^2) of the models under consideration.
- **Bonus Tasks:**
  - **Evaluate Economic Value:** Analyze the economic value of your models through long-short spread portfolios.
  - **Advanced Feature Selection:** Incorporate macroeconomic indicators and use Group LASSO to identify key characteristics.
- Due on 28th Feb

Q&A

- Questions or thoughts about today's lecture?

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